Minsi Hu

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Education

University of Maryland, College Park (UMD) **GPA: 3.99 / 4.00**

- B.S. Computer Science (Machine Learning Concentration) **Received May 2025**
- B.S. Mathematics (Applied Math Concentration)
- M.S. Computer Science **Expected May 2026**

Relevant Coursework

- Machine Learning | Deep Learning | Computer Vision | Natural Language Processing | Multimodal Foundation Models | Computer Graphics | Cryptography | Graph Theory | Real Analysis | Numerical Optimization | Probability and Statistical Theory | Computational Geometry

Experience

Software Engineering Intern | *Capital One, McLean* **June 2025 – August 2025**

- Developed an enterprise-grade **AI chatbot** with **Llama** and **LangChain/LangGraph** to automate data onboarding for fraud detection, collections, and customer service use cases. **Streamlined manual workflows** to accelerate use case delivery and free up the capacity of core team engineers.

Course Instructor and Teaching Assistant | *University of Maryland* **August 2023 – Present**

- Instructor and facilitator for **CMSC389E**: Digital Logic Design Through Minecraft**.**
- Teaching Assistant for the **CMSC132**: Object-Oriented Programming, **CMSC216**: Introduction to Computer Systems**, CMSC330**: Organization of Programming Languages, and **STAT410**: Introduction to Probability Theory courses.
- Taught discussion sections, held office hours, and designed coursework for over **3600 students**.

Undergraduate Research Assistant | *University of Maryland* **December 2023 – Present**

- Previously explored **feature extraction** and **learning-based methods** for processing the human brain's **electroencephalography (EEG)** data with the Neuromotor Control and Learning Laboratory (NMCL).
- Currently working with **Dr. Jia-Bin Huang** in the UMD Vision and Learning Lab on generative models. Previously focused on **multi-subject personalization** in **text-to-image synthesis** (CVPR 2026). Now developing agentic planning systems for educational video generation, targeting NeurIPS 2026.

Personal Projects

DeepCore | **C++ & CUDA** **May 2024 – June 2024**

- Engineered a custom **C++ neural network framework** from scratch, leveraging **NVIDIA's CUDA** platform to accelerate **tensor** operations with **parallel computing** on the GPU. Evaluated to recognize handwritten digits from the MNIST database with over **98% accuracy**.

Violin Fingering Generation | **Python** **April 2024 – April 2024**

- Iteratively developed, trained, and fine-tuned a **novel transformer-based architecture** for autoregressively **predicting optimal violin fingerings** curated to **user preferences** (e.g. skill level, performance style). Utilized **LoRA + prefix** tuning for user personalization. Demonstrated the model's ability to generate ergonomic fingerings for violin passages, verified by a professional musician.

Skills

Languages | Python, C++, C, Java, Rust, OCaml, R, Javascript, Typescript, SQL, C#
Other | CUDA, AWS, PyTorch, HTML, CSS, React, MATLAB, Git, LaTeX

Awards

Grant Family Outstanding Achievement Award in Computer Science and Mathematics **May 2025**
Capital One Bank Dean's Scholarship in Computer Science **January 2025**